MOBILE APP FUNNEL COLLAPSE DETECTOR

Product Analytics — End-to-End Case Study

Total Users	Total Events	A/B Lift	North Star
50,000	**200K+**	**+27.7%**	**0.607**
Simulated	**Across 90 days**	**Statistically sig.**	**Spearman r**

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01 Project Overview

This end-to-end product analytics project simulates a real-world UPI-based fintech app (similar to PhonePe or CRED) with a core feature called **Bill Payment Automation** — where users can set up recurring auto-payments. Despite being live for 6 months, only ~10% of users activate this feature. This project identifies **exactly where users drop off**, validates a **UI redesign A/B experiment**, and defines the **North Star Metric** that best predicts long-term 30-day retention.

Business Questions Answered

- Which funnel step has the highest drop-off and why?
- Did the new UI redesign significantly improve activation rates?
- Which single in-app event best predicts 30-day user retention?
- Which acquisition channel delivers the highest-quality users?

02 Dataset & Tech Stack

Synthetic Dataset

50,000 users over 90 days (Jan–Mar 2024)

200,000+ event-level rows

8 funnel steps per user journey

50/50 A/B experiment split

5 acquisition channels tracked

3 city tiers (Tier 1 / 2 / 3)

Android & iOS device types

Tech Stack

Python 3.10 — data generation & analysis

Pandas & NumPy — data manipulation

SQLite — funnel CTEs

Statsmodels — two-proportion z-test

Scipy — SRM chi-square detection

Matplotlib & Seaborn — visualisations

Tableau Public — interactive dashboard

03 Funnel Analysis

Finding 1 — Step 2 is the Funnel Crisis Point

Address verification (step_2_completed) converts only **42.9%** of users who completed Step 1 — the single largest drop-off in the entire funnel. All other steps convert at 65–92%. Drop-off is **consistent across all device types** (Android 42.8% vs iOS 43.1%) and **all city tiers** (Tier 1: 42.9%, Tier 2: 42.7%, Tier 3: 43.3%), confirming this is a core product design problem — not a network or device-specific issue.

Step-by-Step Funnel Metrics

Funnel Step	Users	Step Conv. %	Status
app_install	49,988	—	✓ Entry
onboarding_started	45,919	91.9%	✓ Good
step_1_completed	38,964	84.8%	✓ Good
step_2_completed	**16,726**	**42.9%**	■ Critical Drop
step_3_completed	13,335	79.7%	✓ Good
bill_linked	10,068	75.5%	✓ Good
automation_activated	5,064	50.3%	■ Watch
first_payment_executed	4,089	80.8%	✓ Good

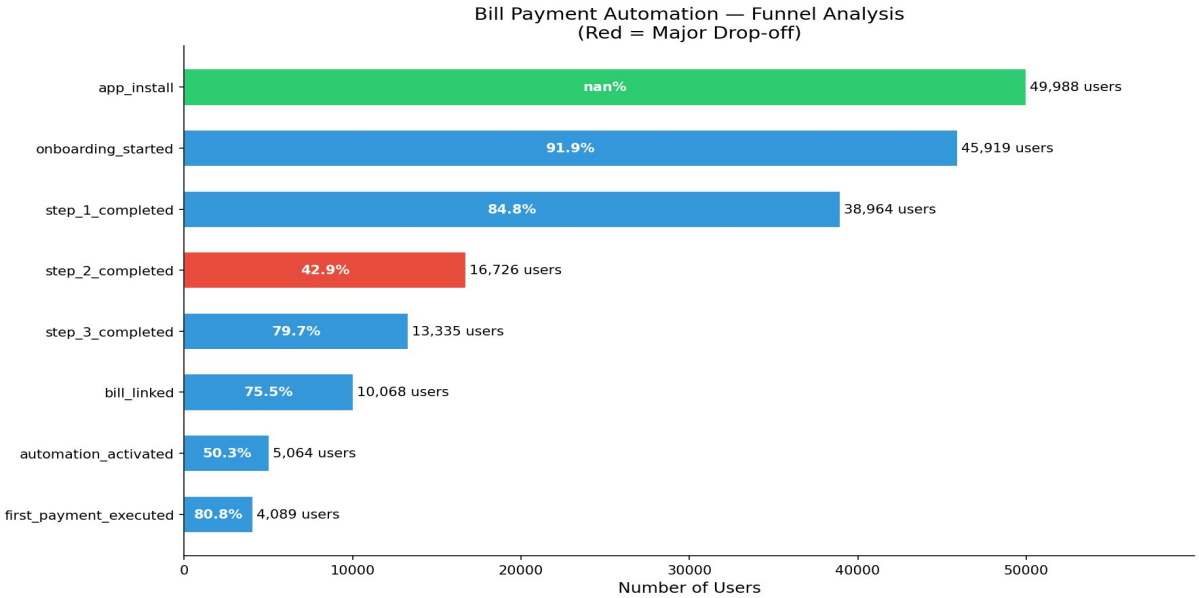

Figure 1 — Funnel chart showing user drop-off at each step. Red bar = critical drop-off point.

04 A/B Test Analysis

Finding 2 — New UI Significantly Improves Activation — Ship It

The treatment group (new UI) achieved an activation rate of **11.37%** vs **8.90%** for control — a **+27.7% relative lift**. The result is statistically significant (p < 0.000001) with a 95% CI of [1.94%, 2.99%] that does not cross zero. The Sample Ratio Mismatch check passed (p=0.65), confirming the experiment is valid. **Recommendation: Ship immediately to 100% of users.**

Metric	Control (Old UI)	Treatment (New UI)
Users	25,044	24,944
Activated Users	2,229	2,836
Activation Rate	8.90%	11.37%
Relative Lift	—	+27.7%
Absolute Difference	—	+2.47 pp
95% Confidence Int.	—	[1.94%, 2.99%]
Z-statistic	—	9.13
P-value	—	< 0.000001
SRM Check	—	PASSED (p=0.65)
Decision	**—**	**SHIP ✓**

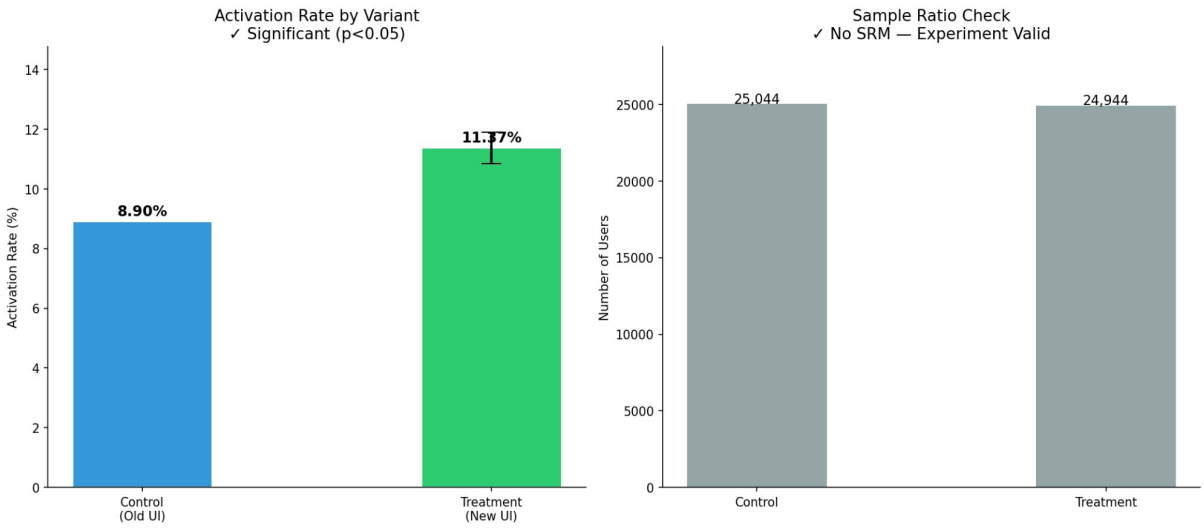

Figure 2 — Left: Activation rates with 95% CI error bar on treatment. Right: SRM check confirming equal group sizes.

05 Cohort Retention Analysis

Users were grouped by their signup week and tracked across four retention windows: Day 1, Day 7, Day 14, and Day 30. The heatmap below shows how retention evolves across 13 weekly cohorts.

Cohort	Size	Day 1	Day 7	Day 14	Day 30
2024-W01 (Jan 1)	3,844	43.9%	20.8%	5.8%	6.3%
2024-W02 (Jan 8)	3,820	44.5%	20.6%	6.9%	6.7%
2024-W03	3,897	45.5%	20.5%	6.8%	6.5%
2024-W08 (Feb 19)	4,006	45.1%	19.4%	6.1%	5.9%
2024-W13 (Mar 25)	3,397	36.7%	2.4%	0.0%	0.0%
Average	**~3,800**	**43.6%**	**19.1%**	**5.3%**	**4.2%**

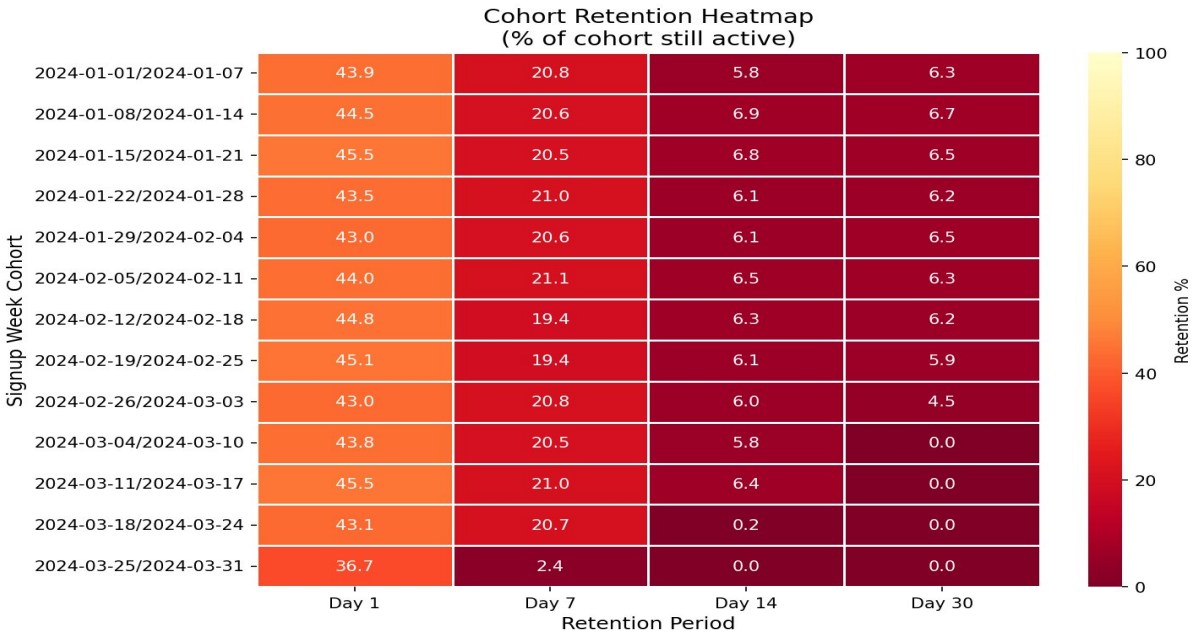

Figure 3 — Cohort retention heatmap. Later cohorts (March) show 0% at Day 30 because they haven't reached Day 30 within the 90-day data window — expected behaviour.

Key Insight:

Retention drops sharply from Day 1 (~44%) to Day 7 (~19%) — a 55% relative decline. This suggests the product is not creating enough habit-forming moments in the first week. The North Star Metric analysis on the next page directly addresses this.

06 North Star Metric Identification

Finding 3 — North Star Metric — automation_activated

Among all tracked in-app events, completing **automation_activated** within the first 7 days has the strongest Spearman correlation (**r = 0.607**, p < 0.0001) with Day 30 retention. This is the single best leading indicator of long-term user value. Users who activate bill automation early are significantly more likely to remain active 30 days later. No other event comes close.

Event	Spearman r	P-value	Significant
automation_activated	**0.6067**	**< 0.0001**	**YES ★**
first_payment_executed	0.5192	< 0.0001	YES
bill_linked	0.4179	< 0.0001	YES
step_3_completed	0.3485	< 0.0001	YES
step_2_completed	0.2964	< 0.0001	YES
step_1_completed	0.1133	< 0.0001	YES
app_open	0.0716	< 0.0001	YES
onboarding_started	0.0653	< 0.0001	YES

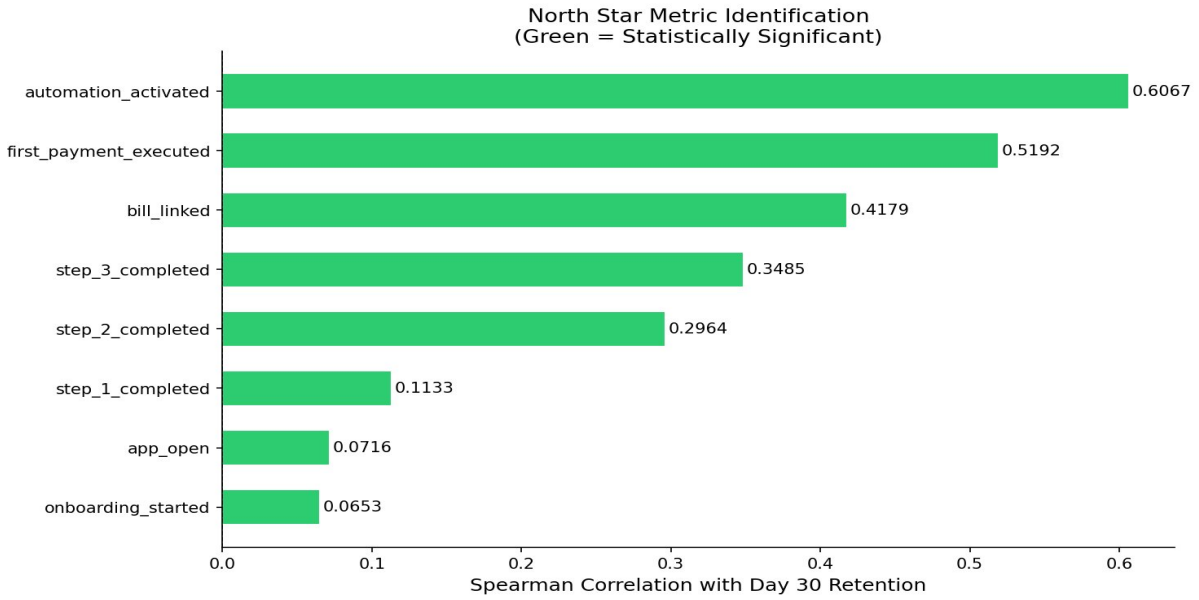

Figure 4 — Spearman correlation of each event with Day 30 retention. automation_activated (r=0.607) is the clear North Star Metric.

07 Recommendations

P0 — Immediate

Fix Step 2 — Aadhaar Auto-fill Integration

Integrate DigiLocker/Aadhaar API at Step 2 to eliminate manual address entry entirely. Users should be able to verify in one tap. Expected impact: reduce Step 2 drop-off from 57% to below 30%, unlocking ~17,000 additional users per 50,000 installs.

P0 — Immediate

Ship the New UI to 100% of Users

A/B test result is conclusive ($p < 0.000001$, SRM passed). Roll out treatment UI immediately. Schedule a 30-day post-launch measurement to confirm results at full scale. Monitor for segment-level degradation (iOS vs Android, Tier 1 vs Tier 3). Expected impact: +27.7% activation rate improvement.

P1 — Next Sprint

Redesign Onboarding to Drive automation_activated in 72 Hours

Restructure Day 0, Day 1, and Day 3 push notifications to guide users toward activating their first automated bill payment within 72 hours of signup. Add automation_activated as the primary KPI on the product team's weekly dashboard, replacing raw install count. Expected impact: measurable improvement in Day 30 retention across all cohorts.

08 Skills Demonstrated

Skill	What Was Done
SQL Funnel Analysis	CTE-based queries in SQLite for step conversion rates
A/B Testing	Two-proportion z-test with p-value and 95% CI
SRM Detection	Chi-square test on group sizes — production-level concept
Cohort Retention	Weekly cohort heatmap across Day 1/7/14/30
North Star Metric	Spearman rank correlation for metric selection
Python Project Structure	Modular src/ + notebooks/ + sql/ layout
Tableau Dashboard	4-sheet interactive public dashboard
Fintech Domain Knowledge	UPI, KYC, bill automation, Indian market context
Executive Communication	Findings translated into PM-ready recommendations

Thank You

This project demonstrates end-to-end product analytics capabilities including funnel analysis, statistically rigorous A/B testing with SRM detection, cohort retention analysis, and North Star Metric identification using Spearman correlation — the exact work product analysts do daily at CRED, Razorpay, PhonePe, and Groww.

GitHub github.com/sabarivarun	**Tableau Dashboard** public.tableau.com/app/profile/sabari.varun	**Analyst** Sabari Varun | April 2024